

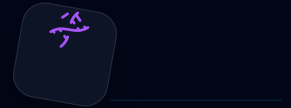


GRADUATION PROJECT 2026

NABD

Intelligent Healthcare Ecosystem

SECURE . RELIABLE . INTELLIGENT_



UNDER SUPERVISION OF

 **Dr. Ahmed Abd-Elmotelb**

OUR TEAM

The Architects of Nabd



Seif El-Din Mohamed

TEAM LEADER & BACKEND



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AI ENGINEER



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FRONTEND DEVELOPER



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UI/UX DESIGNER



Nour El-Din El-Sayed

MOBILE DEVELOPER

NABD

COMPETITIVE ADVANTAGE

WHY NABD?



Intelligent Medical Decision Support

The core strength of our project lies in a high-accuracy AI diagnostic model that maps structured symptom data to diseases. Unlike generic chatbots, our system is **Deterministic** and clinically focused.

95%

TOP-1 ACCURACY

99%

TOP-3 ACCURACY



Unified Medical Passport

A centralized, secure cloud record for every patient. Eliminates fragmented history and ensures doctors always have the full picture.

Full-Cycle Ecosystem



A complete seamless integration replacing isolated apps.

Diag → Book → Pay → Treat

The Core Challenges



Unverified Practitioners

Patients face significant risks with unlicensed doctors and "underground" clinics flooding social media.

- ✗ Lack of trust in digital booking.
- ✗ High probability of fraud.



Random Triage Chaos

Without guidance, patients often book the wrong specialist (e.g., Orthopedist instead of Neurologist).

- ✗ Wasted financial resources.
- ✗ Critical diagnostic delays.



Fragmented History

Medical records are scattered across paper prescriptions that get lost. Doctors treat patients without seeing the full picture.



Operational Inefficiency

Manual booking systems lead to high no-show rates, scheduling conflicts, and crowded waiting rooms.

INTELLIGENT SOLUTIONS

Architecting the Future



Advanced Search & Filter

A smart discovery engine allowing patients to find the perfect match based on location and rating.

- ✓ Precise Specialty.
- ✓ Verified Real Patient Reviews.



Digital Clinic Management

Automated CMS for doctors that eliminates double-booking, manages patient queues, and digitizes records instantly.



Unified Medical Passport

A secure, cloud-based longitudinal record for every patient accessible by authorized doctors only.



AI Diagnostic Assistant

A powerful Deep Learning model supporting clinical decisions by analyzing structured symptoms.

⚡ **99% Top-3** Accuracy.

The Patient Journey



Login

Secure Access



Profile

History & Allergies



Search

Find Verified Doctor



Booking

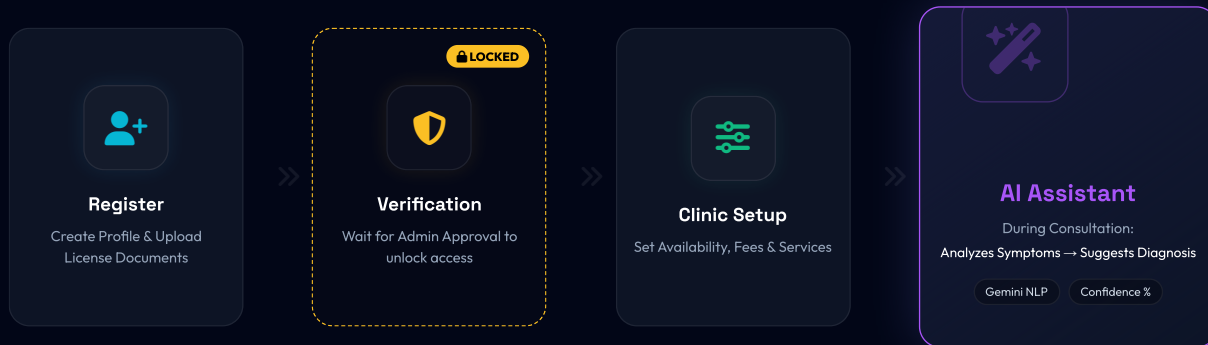
Paymob Payment



Care

Receive E-Rx

The Doctor's Intelligent Workflow



The Verifier (Gatekeeper)

NABD ADMIN

 **Requests** 2

 Verified List

 Statistics

Pending Verification Requests



Dr. Karim Adel

Dermatology Specialist

PENDING REVIEW



Medical_License.pdf



National_ID.jpg

Reject

Verify & Unlock

 **Security Protocol:** Doctors cannot access patient data without this manual verification.

The Diagnosis Challenge

Dataset Composition

377

SYMPTOMS

222

DISEASES

⚠ Critical Challenges:

- **Class Imbalance:** Rare diseases have few samples.
- **Symptom Overlap:** Similar inputs for different conditions.

Preprocessing Pipeline

1. **Cleaning:** Text Normalization
2. **Filtering:** Min 15 samples/class
3. **Encoding:** Binary Vectorization
4. **Balancing:** Weighted Loss Function
5. **Splitting:** Stratified Train/Val

NEURAL NETWORK DESIGN

Deep Learning Architecture

A progressive "Funnel" design for Feature Abstraction & Decision Compression.



Total Parameters: 654,174 (Lightweight & Efficient)

Intelligent Training Strategy



Adam Optimizer

Learning Rate: **0.0005**. Chosen for its adaptive learning capabilities, handling sparse gradients in large feature sets.



Smart Callbacks

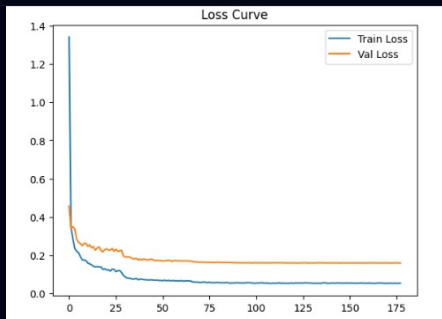
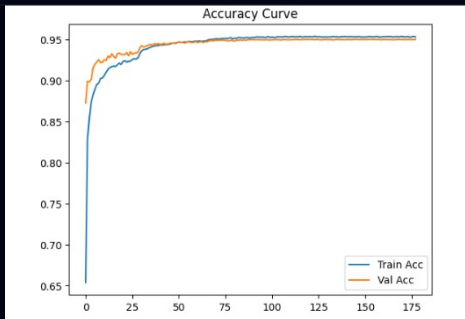
- **EarlyStopping**: Prevents overfitting.
- **ModelCheckpoint**: Saves best weights.
- **ReduceLROnPlateau**: Fine-tunes convergence.



Balanced Weights

Applied `class_weight='balanced'` to penalize mistakes on rare diseases more heavily, ensuring unbiased predictions.

Model Performance



95%

VALIDATION ACCURACY

99%

TOP-3 ACCURACY

Confusion Matrix Insight

"Errors are mostly **semantic** (between similar diseases), with no specific class consistently misclassified."

TERM 2 ROADMAP

NABD

MODEL INTELLIGENCE UPGRADE



LLM Intermediary Bridge

Integrating a powerful language model layer to seamlessly connect raw patient data with our structured diagnostic engine.



Full Medical Narrative

Evolving the model to output comprehensive disease profiles: Detailed Descriptions, Symptom Breakdowns, and Treatment Protocols.

ECOSYSTEM EXPANSION

Native Mobile Experience

Launching a high-performance Flutter application for iOS & Android, putting healthcare control in the patient's pocket.



READY TO DEPLOY?

START PROTOTYPE →



**”Nabd is not just an application,
it is the future of digital
healthcare in Egypt.”**

PRECISION DATA • SMARTER DECISIONS • SAVING LIVES



READY FOR EVALUATION

Experience the Live Prototype